

# Enhancing Customer Profitability and Engagement

## A Data-Driven Analysis of Retail Sales Performance

to conduct a customer profitability analysis

Customers

by: Industry,  
subscription tier,  
region



**Calculate revenue per customer**

Use MRR (Monthly Recurring Revenue) or ARR (Annual Recurring Revenue)

to serve

costs spent on  
volume, and



**Compare revenue vs. costs**

Use a profitability matrix to categorize customers as high-value, low-value, or high-cost

**Take strategic action**

For high-value customers: Focus on retention and expansion

For low-profit customers: Automate support and evaluate retention feasibility.

For mid-tier customers: Look for upsell and cross-sell opportunities



## Executive Summary: The Core Insight

Revenue is concentrated in specific demographics, categories, and locations, indicating a clear path for targeted optimization.

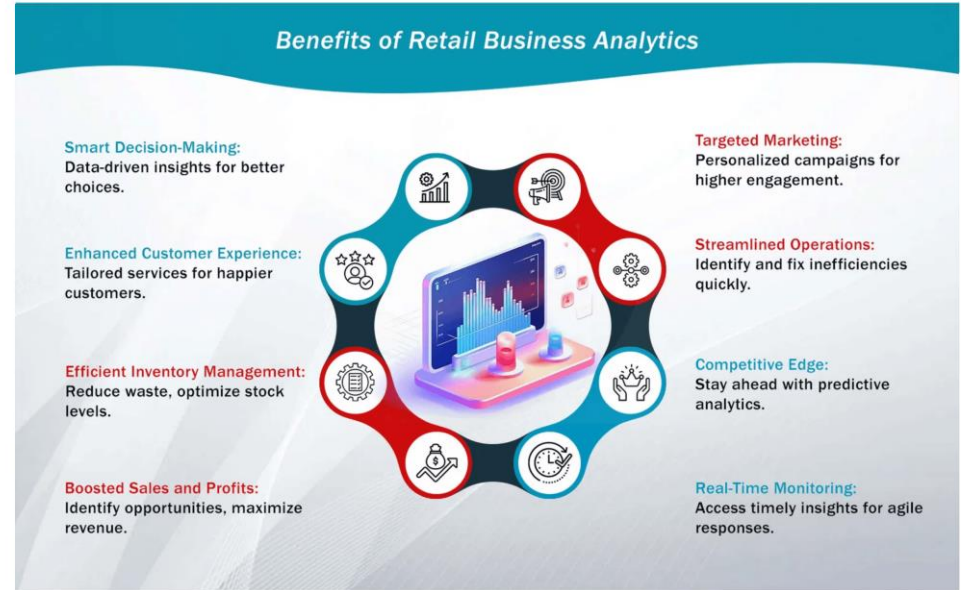
### Problem Statement

- Stagnant revenue growth and lack of personalized customer engagement

### Key Findings

- **Financial Core:** Clothing, Shoes, and Technology drive the majority of revenue
- **Most Valuable Customers:** The 36-50 and 51+ age groups are the highest revenue segments
- **Geographic Outperformers:** Kanyon and Mall of Istanbul are significant revenue drivers

**Strategic Focus:** A three-pronged strategy: Targeted Loyalty, AOV Increase, and Inventory Optimization



# The Business Challenge

**Objective:** Act as a Business Analyst to analyze customer shopping data and provide actionable recommendations for improving revenue growth and customer engagement.

## Core Business Questions Addressed

1. What are the key performance indicators (KPIs) for the business, including total revenue, total items sold, and average transaction value?
2. Which product categories are the primary drivers of revenue and sales volume? Conversely, which categories are underperforming?
3. What is the demographic profile (age and gender) of the customer base, and how does purchasing behavior differ across these segments?
4. Which shopping malls are the top performers versus the underperformers in terms of revenue and transaction volume?
5. What are the most popular payment methods, and does this preference vary by age group?
6. Based on all the above insights, what are the top 3-5 actionable recommendations to improve customer retention, increase average order value, and optimize inventory?



## Methodology: A Multi-Tool Approach

### 1 Data Warehousing & Enrichment

Google BigQuery - SQL

- Cleaned raw data and verified data types
- Engineered new features (total\_price, age\_group)

### 2 Exploratory Data Analysis

Python - Pandas, Matplotlib, Seaborn

- Conducted deep exploratory analysis to uncover trends
- Generated detailed static visualizations for hypothesis testing

### 3 Interactive Dashboarding

Microsoft Power BI

- Connected directly to cleaned BigQuery table
- Created fully interactive dashboard for business user exploration

# Finding 1: High-Level Performance Snapshot

251.5M

Total Revenue

299K

Total Items Sold

2.53K

Average Transaction Value (AOV)

28-29M

Quarterly Revenue (2021-2022)

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**Revenue Consistency:** Quarterly revenue remained stable at approximately 28-29M from 2021 through 2022, demonstrating consistent performance and strong revenue fundamentals. This stability indicates a mature, well-established business with predictable cash flows and customer loyalty.

## Finding 2: Category Dominance

Clothing, Shoes, and Technology form the financial core of the business.

### Top Revenue Categories

- **Clothing:**114M
- **Shoes:**67M
- **Technology:**58M

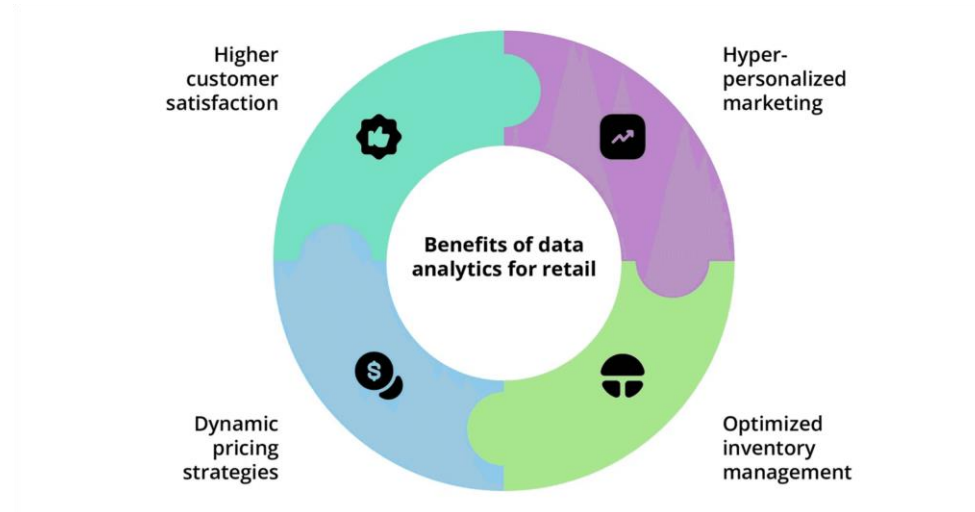
### Top Sales Volume Categories

- Clothing, Cosmetics, and Food & Beverages account for **65% of total items sold**

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### Underperforming Categories

Books, Souvenirs, and Toys show significantly lower revenue and sales volume, indicating reduced customer demand and lower profitability in these categories.



## Finding 3: Demographic Value

Older demographics are the most valuable customer segments, driving the highest revenue.

### Highest Revenue Segments

- Age Group **36-50**
- Age Group **51+**

### Gender Distribution

- **Females:** Approximately **60% of sales and revenue**, indicating stronger purchasing power among female customers
- **Males:** Approximately 40% of sales and revenue

### Payment Method Preferences

- **Cash:** Most used method at **45%** of total payments
- **Credit & Debit Cards:** Widely used alternative payment methods
- **Consistency:** Payment preference remains consistent across all age groups



# Finding 4: Geographic Concentration

Revenue is highly concentrated in two key locations, indicating strong market presence and customer loyalty in these specific areas.

## Significant Outperformers

### Kanyon Mall

A significant revenue driver with disproportionately large share of total revenue and sales volume

### Mall of Istanbul

A significant revenue driver with disproportionately large share of total revenue and sales volume

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## Actionable Implication

These malls indicate strong market presence and customer loyalty, justifying increased investment in localized marketing strategies, inventory optimization, and customer engagement initiatives tailored to each location's unique customer base.



# Strategic Recommendation 1: Customer Retention

Focus: Capitalize on Top Performers

## Action 1: Loyalty Program for Older Demographics

- Launch a tiered loyalty program specifically marketed to the 36-50 and 51+ age groups with exclusive benefits and rewards for repeat purchases

### Expected Impact

- Increased Customer Lifetime Value (CLV) from the most profitable demographic
- Reduced churn and improved customer loyalty

## Action 2: Personalized Marketing Campaigns

- Implement targeted emails and app notifications for female customers (60% of sales) with promotions on Clothing and Cosmetics

### Expected Impact

- Higher engagement and conversion rates from marketing campaigns
- Increased sales in top-performing categories (Clothing and Cosmetics)



## Strategic Recommendation 2: Increase Average Order Value

**Focus:** Drive Higher Transaction Value through strategic bundling and promotional tactics.

### Action 1: Product Bundling and Cross-Selling

- Implement product bundling and cross-selling featuring items from Clothing, Shoes, and Technology (e.g., "complete the look" bundles with matching clothing and shoes, or "tech essentials" bundles)

#### Expected Impact:

- Direct increase in average transaction value
- Increased sales of items from core business categories
- Improved customer experience through convenient, curated offerings

### Action 2: "Spend and Save" Promotions

- Introduce tiered in-store offers with thresholds set just above the current AOV of 2.53K (e.g., "Spend 3.0K, get 10% off" or "Buy 2 items from Clothing category, get the 3rd 50% off")

#### Expected Impact:

- Significant lift in average spend per customer
- Encourages customers to purchase additional items to reach promotional thresholds
- Increases overall revenue while maintaining healthy profit margins



## Strategic Recommendation 3: Inventory Optimization

**Focus:** Align stock levels with category performance to maximize revenue and reduce carrying costs.

### Action 1: Increase Inventory for High-Performing Categories

- Increase inventory for Clothing, Shoes, and Technology to avoid stockouts and ensure customer satisfaction

#### Expected Impact

- Increased sales and customer satisfaction by ensuring availability of popular products
- Reduced lost sales due to stockouts in high-demand categories

### Action 2: Reduce Stock for Underperforming Categories

- Reduce inventory and potentially delist underperforming categories (Books, Souvenirs, Toys)

#### Expected Impact

- Reduced inventory holding costs and warehouse management expenses
- Improved cash flow by investing capital in high-return inventory
- Simplified product portfolio and improved inventory turnover



# Conclusion & Next Steps

## Project Impact

### 1. Clarity:

Defined the financial core of the business across categories, demographics, and geography, providing a clear understanding of revenue drivers and customer value segments.

### 2. Strategy:

Provided five concrete, data-driven recommendations across customer retention, average order value, and inventory optimization with quantified expected business impacts.

### 3. Actionability:

Delivered specific, implementable initiatives that can be executed immediately with clear metrics for success and ROI measurement.

## Recommended Next Steps

### 1. Implement Loyalty Program Pilot:

Launch the tiered loyalty program in Kanyon and Mall of Istanbul (top-performing locations) with the 36-50 and 51+ age groups as primary targets. Measure CLV increase and churn reduction.

### 2. A/B Test "Spend and Save" Promotions:

Test different promotional thresholds (3.0K, 3.5K, 4.0K) and discount levels (10%, 15%, 20%) to identify the optimal balance between AOV lift and margin preservation.

### 3. Establish Quarterly Review Cycle:

Create a quarterly review process to monitor category performance, adjust inventory levels, and track the effectiveness of all implemented recommendations against baseline metrics.